is one performed by Michelson and Morley in 1887. It was 18 years later before the negative results of the experiment were finally explained, by Einstein.

The Michelson-Morley experiment was performed with an apparatus like that shown schematically in Figure 3-2. This apparatus is essentially comprised of a light source *A*, a partially silvered glass plate *B*, and two mirrors *C* and *E*, all mounted on a rigid base. The mirrors are placed at equal distances *L* from *B*. The plate *B* splits an oncoming beam of light, and the two resulting beams continue in mutually perpendicular directions to the mirrors, where they are reflected back to *B*. On arriving back at *B*, the two beams are recombined as two superposed beams, *D* and *F*. If the time taken for the light to go from *B* to *E* and back is the same as the time from *B* to *C* and back, the emerging beams *D* and *F* will be in phase and will reinforce each other, but if the two times differ slightly, the beams will be slightly out of phase and interference will result. If the apparatus is "at rest" in the ether, the times should be precisely equal, but if it is moving toward the right with a velocity *u*, there should be a difference in the times. Let us see why.

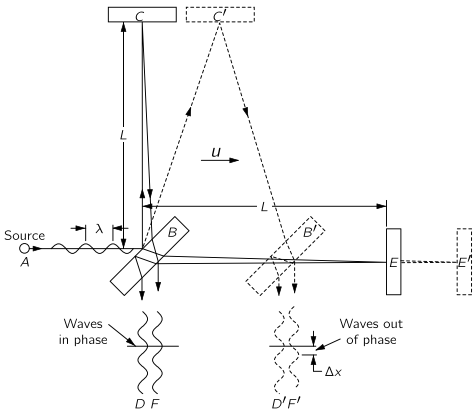


Figure 3-2 Schematic diagram of the Michelson-Morley experiment.